

Research Paper Revision Summary

Project: Smart Civic Issue Register & Tracker using Geospatial Analytics and Machine Learning

The attached manuscript is a research-paper extension of the EPICS project previously completed under your supervision. The work has been substantially expanded beyond the original EPICS implementation to improve its research depth, validation methodology, and experimental analysis.

Major Enhancements

1. Expanded Machine Learning Evaluation

- Added Random Forest and XGBoost models alongside the original baseline classifiers.
- Performed comparative evaluation using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics.
- Added statistical significance testing to support performance comparisons.

2. Spatial Feature Engineering Analysis

- Evaluated the contribution of density-based spatial features through controlled ablation studies.
- Quantified the impact of density information on civic issue prioritization performance.
- Investigated the interaction between spatial density and geographic coordinates.

3. Real-World Validation

- Introduced NYC 311 civic complaint data as an external validation dataset.
- Conducted direct training experiments on real-world complaint records.
- Performed synthetic-to-real transfer evaluation to study domain-shift effects.

4. Hotspot Detection and Spatial Analytics

- Extended DBSCAN-based hotspot analysis.
- Added real-world hotspot evaluation using NYC complaint data.
- Compared spatial patterns across controlled and real-world environments.

5. Robustness and Feature Analysis

- Performed feature importance analysis.
- Evaluated model behavior after removing status-related and spatial features.
- Added discussion on dataset dependency and transferability of spatial signals.

Key Findings

- Spatial density features provide meaningful improvements in the controlled synthetic environment.
- Real-world performance is influenced more strongly by complaint status and administrative indicators.
- Spatial information remains useful but behaves differently across datasets.

- Domain shift significantly impacts direct transfer between cities, highlighting the importance of local calibration.

Current Status

- IEEE-format conference manuscript completed.
- All reviewer comments addressed.
- Real-world validation incorporated.
- References verified and updated.
- Final manuscript compiled successfully without citation or formatting issues.

I would be grateful for any suggestions regarding the technical content, presentation quality, and overall publication readiness of the manuscript.